Woo Kyung Kim

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Education

SungKyunKwan University — Gyeonggi-do, Korea
Ph.D in Computer Science and Engineering — *Sep. 2022 – Current*
- Major GPA:4.15/4.5
- Advisor: Prof. Honguk Woo

SungKyunKwan University — Gyeonggi-do, Korea
Master of Computer Science and Engineering — *Mar. 2021 – Aug. 2022*
- Major GPA:4.5/4.5
- Advisor: Prof. Honguk Woo

SungKyunKwan University — Gyeonggi-do, Korea
Bachelor of Computer Science and Engineering — *Mar. 2017 – Feb. 2021*
- Major GPA:4.41/4.5, Total GPA:4.26/4.5

Korean Minjok Leadership Academy — Gangwon-do, Korea
High School — *Mar. 2014 – Feb. 2017*

Publications

[RL]: Reinforcement Learning; [SL]: Skill-based Policy Learning; [EP]: Embodied Planning; [AA]: Agentic AI;

[c.13] [SL] Skill-Adaptive Noise Scheduling for Diffusion Policies
Woo Kyung Kim, Gwangpyo Yoo, Eunyoung Park, Honguk Woo
Conference on Neural Information Processing Systems (NeurIPS), 2026.12

[c.12] [AA] Code-as-Auditor: Executable Compliance Reasoning via Regulation-to-Code
Jisoo Kim, Taeyoon Kwack, Jinwoo Jang, **Woo Kyung Kim**, Honguk Woo
ACM International Conference on Information and Knowledge Management (CIKM), 2026.11

[c.11] [EP] Amortizing Self-Correction for Continual Embodied Adaptation
Woo Kyung Kim, Jooyoung Kim, Minjong Yoo, Honguk Woo
Asia-Pacific Chapter of the Association for Computational Linguistics (AACL), 2026.11

[c.10] [AA] MA-Mesh: Resilient Multi-Agent Collaboration via Role-based Self-Healing
Junyong Lee*, **Woo Kyung Kim***, Honguk Woo
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.10

[c.09] [RL] Knothe-Rosenblatt Quantile Regression for Risk-sensitive Multi-objective Reinforcement Learning
Gwangpyo Yoo, **Woo Kyung Kim**, Honguk Woo
International Conference on Machine Learning (ICML), 2026.07

[j.02] [AA] Aspect-Augmented Distillation of Task-Oriented Dialogues to Small Language Models
Jongmoon Jun, **Woo Kyung Kim**, Hyunseong Na, Honguk Woo, Jeehyeong Kim
Expert Systems with Applications, vol. 302, pp. 130494, 2025.11

[c.08] [SL] In-context Policy Adaptation via Cross-domain Skill Diffusion
Minjong Yoo, **Woo Kyung Kim**, Honguk Woo
AAAI Conference on Artificial Intelligence (AAAI), 2025.02

[c.07] [SL] LLM-based Skill Diffusion for Zero-shot Policy Adaptation
Woo Kyung Kim, Youngseok Lee, Jooyong Kim, Honguk Woo
Conference on Neural Information Processing Systems (NeurIPS), 2024.12

[c.06] [SL] Incremental Learning of Retrievable Skills for Efficient Continual Task Adaptation
Daehee Lee, Minjong Yoo, **Woo Kyung Kim**, Wonje Choi, Honguk Woo
Conference on Neural Information Processing Systems (NeurIPS), 2024.12

[c.05] [RL] Pareto Inverse Reinforcement Learning for Diverse Expert Policy Generation
Woo Kyung Kim, Minjong Yoo, Honguk Woo
International Joint Conference on Artificial Intelligence (IJCAI), 2024.08

[**c.04**] [EP] Embodied CoT Distillation From LLM to Off-the-shelf Agents
Wonje Choi, **Woo Kyung Kim**, Minjong Yoo, Honguk Woo
International Conference on Machine Learning (ICML), 2024.07

[**c.03**] [SL] Robust Policy Learning via Offline Skill Diffusion
Woo Kyung Kim, Minjong Yoo, Honguk Woo
AAAI Conference on Artificial Intelligence (AAAI), 2024.02

[**c.02**] [EP] Efficient Policy Adaptation with Contrastive Prompt Ensemble for Embodied Agents
Wonje Choi, **Woo Kyung Kim**, SeungHyun Kim, Honguk Woo
Conference on Neural Information Processing Systems (NeurIPS), 2023.12

[**c.01**] [SL] One-shot Imitation in a Non-Stationary Environment via Multi-Modal Skill
Sangwoo Shin, Daehee Lee, Minjong Yoo, **Woo Kyung Kim**, Honguk Woo
International Conference on Machine Learning (ICML), 2023.07

[**j.01**] [RL] Repot: Transferable Reinforcement Learning for Quality-Centric Networked Monitoring in Various Environments
Youngseok Lee, **Woo Kyung Kim**, Sung Hyun Choi, Ikjun Yeom, Honguk Woo
IEEE Access, vol. 9, pp. 147280-147294, 2021.11

Experience

CSI Agent Lab Aug. 2026 – Present
Lab Leader; Physical AI Team Leader

- Lead a lab of ∼30 students and mentor a 10-member Physical AI team

SKKU Student Ambassador Mar. 2017 – Aug. 2018
President

- Promoted the computer science and engineering department to high school students through nationwide tours.

Awards

AI Co-Scientist Challenge Korea Jan. 2026 – Apr. 2026

- RAPIDS: Robot AI Agents Pipeline for Intelligent Data Collection & Skill Learning

Skills & Interests

Research Interests: Skill-based Reinforcement Learning, Diffusion Model, Embodied Agent

Robots:
Manipulators: UR7e + Robotiq 2F-85, AgileX Piper, LeRobot SO-101
Dexterous Hands: Inspire RH5DG2 (13-DoF), Tesollo DG-5F (20-DoF)
Humanoids: Rainbow RB-Y1, Unitree G1

Language: Korean (native), English (Fluent), Mandarin Chinese (intermediate)

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